# FloodMax Algorithm: Leader Election in General Graphs

*Synthesis based on Arpit Bhayani's Distributed Systems Masterclass • Lynch (Distributed Algorithms)*

**ALGORITHMIC PREMISE & GENERAL GRAPH CONSENSUS**

Real-world distributed systems form arbitrary mesh graphs rather than symmetric rings. The FloodMax algorithm solves leader election on any connected undirected graph $G = (V, E)$ under synchronous timing assumptions. By leveraging knowledge of the **Network Diameter ($D = ext{diam}(G)$)**, nodes flood the maximum identifier encountered across $D$ rounds. After exactly $D$ rounds, the global maximum $ ext{UID}_{\max}$ is guaranteed to have reached every node, achieving time-optimal termination in $\mathcal{O}(D)$ rounds and $\mathcal{O}(D \cdot |E|)$ messages.

## 1. MOVING BEYOND RINGS: GENERAL GRAPH CHALLENGES

In general topologies, nodes have non-uniform degrees and lack global orientation (no clockwise/counter-clockwise channels). A node has only local visibility into its adjacent neighbors $\mathcal{N}(u)$. The core engineering challenge is defining a deterministic termination condition without centralized coordination.

## 2. SYSTEM MODEL & THE NETWORK DIAMETER (D)

### Network Topography

- **Connected Undirected Graph:** $G = (V, E)$ with $n = |V|$ nodes and $m = |E|$ edges.
- **Synchronous Rounds:** Execution progresses in synchronized lock-step rounds $1, \dots, D$.
- **Unique Total Ordering:** Distinct identifiers $ ext{UID}(u) \in \mathcal{U}$.

### Diameter Definition

The diameter $D = \max_{u, v} ext{dist}(u, v)$ is the maximum shortest path between any vertex pair. Nodes know $D$ (or an upper bound $\Delta \ge D$).

## 3. PROTOCOL SPECIFICATION & WAVEFRONT FLOODING

1. **Initialization:** Each node $u$ sets $ ext{max_uid}_u \gets ext{UID}(u)$.
2. **Round 1:** Node $u$ broadcasts $ ext{max_uid}_u$ to all neighbors $\mathcal{N}(u)$.
3. **Rounds $2 \le r \le D$:**
   - Node $u$ collects identifiers from all adjacent neighbors.
   - Node $u$ updates: $ ext{max_uid}_u \gets \max( ext{max_uid}_u, \max_{v \in \mathcal{N}(u)} M_v)$.
   - Node $u$ broadcasts the updated maximum to all neighbors.
4. **Termination at Round $D$:** If $ ext{max_uid}_u == ext{UID}(u)$, process $u$ becomes `LEADER`; otherwise it becomes `FOLLOWER`.

## 4. FORMAL PROOF OF CORRECTNESS

Let $u^* \in V$ be the unique node with $ ext{UID}(u^*) = \max_{v \in V} ext{UID}(v)$.

**Wavefront Propagation Lemma**

For any round $r \in \{1, \dots, D\}$ and for any vertex $w \in V$:

$$ ext{dist}(u^*, w) \le r \implies ext{max_uid}_w = ext{UID}(u^*) ext{ at the end of round } r$$

**Proof by Induction on Round $r$**

- **Base Case ($r = 1$):** In round 1, $u^*$ broadcasts $ ext{UID}(u^*)$ to all immediate neighbors. Because $ ext{UID}(u^*)$ is the global maximum, every distance-1 neighbor sets its $ ext{max_uid} \gets ext{UID}(u^*)$. Holds.
- **Inductive Step:** Assume the lemma holds for round $r-1$. Let $z$ be any node with $ ext{dist}(u^*, z) = r$. There exists a neighbor $y \in \mathcal{N}(z)$ along a shortest path from $u^*$ with $ ext{dist}(u^*, y) = r-1$. By induction, node $y$ holds $ ext{UID}(u^*)$ at round $r-1$ and transmits it to $z$ in round $r$. Node $z$ updates its maximum to $ ext{UID}(u^*)$. Holds for all $r$.

**Safety and Liveness Theorem**

By definition of diameter, every node in $V$ is at distance at most $D$ from $u^*$. Thus, at round $D$, 100% of nodes in the graph hold $ ext{max_uid} = ext{UID}(u^*)$. Only $u^*$ has $ ext{UID}(u^*) == ext{max_uid}$, guaranteeing that exactly one leader is elected globally.

## 5. COMPLEXITY ANALYSIS ACROSS TOPOLOGIES

**Time Complexity: $\mathcal{O}(D)$ Rounds**

Termination occurs in exactly $D$ rounds. This is strictly **time-optimal**: information from the furthest node requires at least $D$ hops to propagate across the graph.

**Message Complexity: $\mathcal{O}(D \cdot |E|)$**

In each round, each undirected edge carries at most 2 messages (one per direction). Total messages across $D$ rounds: $M \le 2 D \cdot |E| = \mathcal{O}(D \cdot |E|)$.

## 6. TOPOLOGY TRADE-OFFS & SPECIAL GRAPH BOUNDS

| Graph Topology | Diameter $D$ | Edges $\|E\|$ | Rounds to Terminate | Total Messages |
|---|---|---|---|---|
| **Complete Graph (Clique, $K_n$)** | $D = 1$ | $\Theta(n^2)$ | $1$ round (Instantaneous) | $\Theta(n^2)$ |
| **Star Graph ($S_n$)** | $D = 2$ | $n - 1$ | $2$ rounds | $\Theta(n)$ (Linear) |
| **Ring Topology ($C_n$)** | $D = \lfloor n/2 \rfloor$ | $n$ | $\lfloor n/2 \rfloor$ rounds | $\Theta(n^2)$ |
| **Linear Line Graph ($L_n$)** | $D = n - 1$ | $n - 1$ | $n - 1$ rounds | $\Theta(n^2)$ |

## 7. PRODUCTION OPTIMIZATION: FLOODMAX WITH SILENCING

In naive FloodMax, every node transmits in every round regardless of updates. The **Silencing Rule** optimizes this:

**Silencing Condition:** In round $r \ge 2$, a node transmits to its neighbors *if and only if its local $ ext{max}_{uid}$ was strictly updated* in round $r-1$.
In clustered or dense graphs, this prevents redundant transmissions, reducing real-world packet traffic by up to 80% without affecting correctness.

## 8. ARCHITECTURAL ENGINEERING TAKEAWAYS

1. **Diameter Governs Consensus Horizons:** In any distributed network lacking centralized arbiters, the time required to establish uncoordinated consensus is fundamentally lower-bounded by the graph diameter.
2. **Robustness to Graph Perturbations:** Node and link failures can dynamically stretch graph diameter. When deploying FloodMax, configure the round timeout with a conservative upper bound $\Delta \ge D$.
3. **Foundation for Spanning Trees:** The propagation wavefront established during FloodMax naturally traces a Shortest Path Tree from the elected leader, providing the underlying structure for distributed BFS and convergecast operations.